

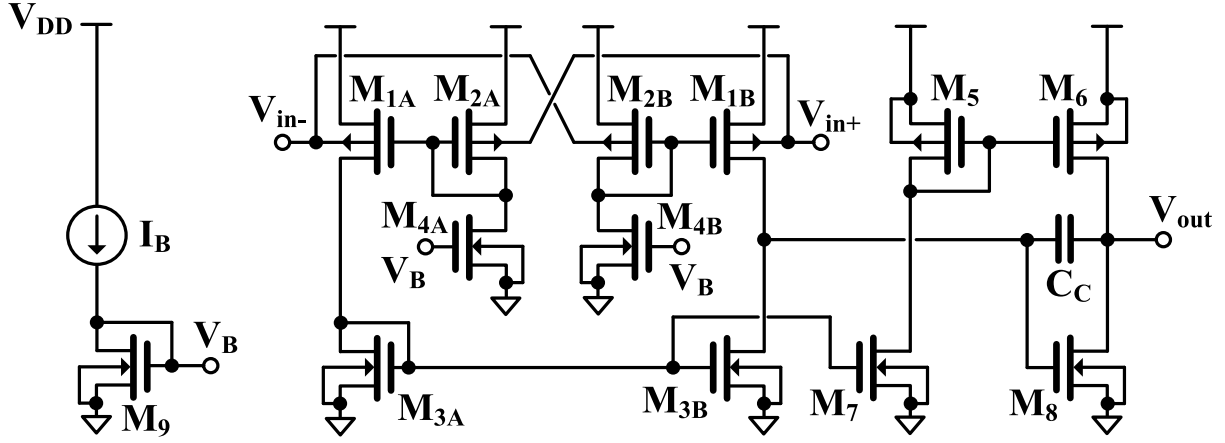


Fig. 1. CMOS schematic of the proposed OTA.

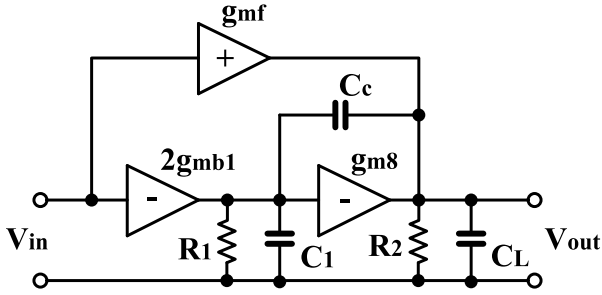


Fig. 2. Block diagram of OTA.

the same time [16], [19], [21]. In addition, the circuit operates in class AB, which increases its SR performance. It is worth noting that its noise properties and the input-referred offset are not worse than achieved for a conventional BD differential pair (see the Appendix).

The second gain stage of the OTA is formed by the transistor  $M_8$ , loaded by the transistor  $M_6$ . The capacitor  $C_C$  is used for frequency compensation. As it is well known, the compensation capacitor introduces a parasitic RHP zero, which reduces the phase margin (PM) and, consequently, limits the GBW product of a two-stage Miller OTA. There are several techniques to mitigate or even completely remove the impact of the RHP zero. In the proposed design, the MMZCC technique [20] has been applied. The block diagram of the OTA with MMZCC is shown in Fig. 2, where  $2g_{mb1}$  stands for the transconductance of the input stage [19],  $g_{m8}$  is the transconductance of the output stage, and  $g_{mf}$  is the transconductance of the feedforward path, which compensates for the direct feedthrough effect associated with  $C_C$ . In the proposed design, this path is created by the input stage and the transistors  $M_5$ – $M_7$ . The resistances  $R_1$  and  $R_2$  represent the output resistances of the input and output stages, respectively, while  $C_1$  and  $C_L$  represent their load capacitances.

#### A. Small-Signal Performance

The transfer function of the MMZCC structure shown in Fig. 2 is given by

$$\frac{V_{out}}{V_{in}} \cong \frac{2g_{mb1}g_{m8}R_1R_2\left[1 + s\frac{(g_{mf}-2g_{mb1})C_C}{2g_{mb1}g_{m8}}\right]}{(1 + sg_{m8}R_2R_1C_C)(1 + sC_L/g_{m8})}. \quad (1)$$

Thus, the transfer function (1) has two left-half-plane (LHP) poles and one zero as follows:

$$p_1 \cong -\frac{1}{g_{m8}R_2R_1C_C} \quad (2)$$

$$p_2 \cong -\frac{g_{m8}}{C_L} \quad (3)$$

$$z_1 \cong -\frac{2g_{mb1}g_{m8}}{(g_{mf} - 2g_{mb1})C_C}. \quad (4)$$

Note that the zero  $z_1$  can be LHP if  $g_{mf} > 2g_{mb1}$ . It is also possible to cancel  $p_2$  with  $z_1$  if the following condition is met:

$$g_{mf} = 2\left(1 + \frac{C_L}{C_C}\right)g_{mb1}. \quad (5)$$

The transconductance  $g_{mf}$  for the considered structure can be expressed as

$$g_{mf} = g_{mb1}\frac{g_{m7}}{g_{m3}} \cdot \frac{g_{m6}}{g_{m5}}. \quad (6)$$

Thus, from (5) and (6), assuming that all transistors operate in the subthreshold region where their transconductances are proportional to the biasing currents, we obtain the following condition for pole-zero cancellation:

$$\frac{I_{D6}}{I_{D3}} = 2\left(1 + \frac{C_L}{C_C}\right). \quad (7)$$

When the above condition is met, the transfer function (1) shows one dominant pole  $p_1$  only. It is worth noting that meeting this condition in ULP circuits may be difficult because a relatively large quiescent current of the output stage ( $I_{D6}$ ) is generally required. The  $I_{D6}/I_{D3}$  ratio may be lowered with increasing  $C_C$ ; however, this leads to lower GBW, SR, and increased silicon area. Thus, the above compensation is useful for small loading capacitances (i.e., small  $C_L/C_C$  ratios achieved for relatively small  $C_C$ ).

In the proposed design, a current push–pull version of the feedforward path  $g_{mf}$  was applied [22], which improves the driving capability of the output stage. The drain current of  $M_3$  is mirrored to the output through the current mirrors  $M_3$ – $M_7$  and  $M_5$ – $M_6$ , thus increasing the current sourced to the load. On the other hand,  $M_8$  provides the sinking current ability. Consequently, a push–pull amplifier is created in a very simple manner.